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MILITARY SPECIFICATION

PISTONS, HYDRAULIC BRAKE, WHEEL CYLINDER:
SINTERED IRON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fluid-impregnated (non-petroleum type), sintered iron pistons used in wheel cylinders of hydraulic brake systems (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Federal

Fed. Test Method Std. No. 791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC-2530

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E18	-	Rockwell Hardness and Rockwell Superfical Hardness of Metallic Materials.
ASTM D92	-	Flash and Fire Points by Cleveland Open Cup.
ASTM D97	-	Pour Point.
ASTM B328	-	Density and Interconnected Porosity of Sintered Powder Metal Structural Parts and Oil-Impregnated Bearings.
ASTM D445	-	Viscosity of Transparent and Opaque Liquids (Kinematic and Dynamic Viscosities).
ASTM D941	-	Density and Specific Gravity of Liquids by Lipkin Bicapillary Pycnometer.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pennsylvania 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC.
SAE Standard J1703 - Motor Vehicle Brake Fluid.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 485 Lexington Avenue, New York, N. Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. A first article of the sintered iron pistons to be furnished under this specification shall be produced prior to the manufacture of this item in production quantity. The first article of the sintered iron pistons and the impregnate shall be submitted to the Government for first article inspection to determine conformance to the quality assurance provisions of this specification (see 4.2). The first article of the sintered iron pistons submitted by a contractor shall be fully representative of the pistons to be supplied from production facilities and tooling. The first article sample of the impregnate shall be representative of the impregnate used in production of these pistons.

3.2 Materials. The piston shall be of sintered iron, lead-tin alloy manufactured from powder composition specified in paragraph 3.3 and will satisfy the physical and performance requirements of this specification. The pistons shall be furnished impregnated with a water insoluble polyoxyalkylene glycol-type synthetic lubricant having additives if necessary to improve the viscosity-temperature characteristics and resistance to oxidation of the impregnate and corrosion resistance of the finished product.

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3.3 Chemical composition. The chemical composition of the powder material shall be as follows:

Lead	7.5 - 10.0%
Tin	0.90 - 1.50%
Carbon	0.20% max
Total other elements	1.0% max
Iron	Balance

3.4 Physical properties.

3.4.1 Piston.

3.4.1.1 Rockwell hardness. When tested as specified in 4.5.1.1, the piston shall have a Rockwell hardness of H76 - H98.

3.4.1.2 Density.

3.4.1.2.1 Unimpregnated piston. When tested as specified in 4.5.1.2, the unimpregnated piston shall have a density of 6.3 - 6.9 grams per cubic centimeter (G/CC).

3.4.1.2.2 Impregnated piston. When tested as specified in 4.5.1.2, the impregnated piston shall have a density of 6.4-7.1 G/CC. The increase in density between the unimpregnated and impregnated piston will be a minimum of 0.1 G/CC.

3.4.1.3 Porosity. When tested as specified in 4.5.1.3, the intercommunicating porosity shall be from 12 - 21 percent.

3.4.1.3.1 Profilograph. When tested as specified in 4.5.1.3.1, the roughness profile shall be composed of plateaus and valleys. Peaks, if present, shall not project more than 50 microinches above the tops of the adjacent plateaus. There is no restriction on the depth of the valleys or voids.

3.4.1.4 Pressure resistance. When tested as specified in 4.5.1.4:

(a) An internal fluid pressure of 3,000 pounds per square inch (psi) for 30 seconds shall not result in any doming of the piston.
(b) An internal fluid pressure of 5,000 pounds per square inch (psi) or cylinder burst pressure (whichever is lower), for 30 seconds shall result in no puncture, penetration or visible cracks.

3.4.1.5 Crush resistance. When tested as specified in 4.5.1.5, the piston shall withstand a radial crush pressure of 5000 pounds minimum. Crush resistance requirement is waived for insert type - shoe loaded sintered iron pistons.

3.4.2 Impregnate.

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3.4.2.1 Viscosity. When tested as specified in 4.5.2.1, the viscosity of the impregnate shall be 550 - 1200 Saybolt Seconds Universal (SSU).

3.4.2.2 Specific gravity. When tested as specified in 4.5.2.2, the specific gravity of the impregnate shall be 0.997 - 1.003.

3.4.2.3 Pour point. When tested as specified in 4.5.2.3, the pour point shall be below minus 15°F.

3.4.2.4 Flash point. When tested as specified in 4.5.2.4, the flash point shall be above plus 420°F.

3.5 Performance (impregnated piston).

3.5.1 Stroking (endurance). When tested as specified in 4.5.3, wear on the pistons shall be less than 0.001 inch. No scoring shall be permitted on the pistons or cylinder walls. There shall be no malfunction of the stroking apparatus which could be attributed to the sintered iron pistons.

3.5.2 Corrosion resistance. After being tested as specified in 4.5.4, there shall be no iron oxide deposits on the pistons or on the portion of the cylinder walls which are in contact with the pistons. Rust deposits in other areas are not cause for failure.

3.6 Workmanship. The pistons shall be uniform in composition and free from scratches, nicks, chips, burrs, and other defects which would affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection. A first article sample consisting of 40 impregnated pistons of which 20 pistons shall be fully assembled in wheel cylinder assemblies with push rods, and one pint of impregnate shall be submitted for first article inspection (see 6.2). The samples shall be representative of the materials proposed to be furnished under the contract. First article inspec-tion shall be conducted by the Government, at a place designated by the procuring activity, and shall consist of examination for the defects specified in 4.3.2.2, and tests as specified in table II.

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4.2.1 Failure. Failure of a first article to pass any examination or test specified herein may be cause, at the option of the Government, for refusal to conduct additional testing until the faults revealed by the test have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. A lot shall consist of all pistons of one part number, representative of production. Selection shall be at random. One day's production shall be considered a lot.

4.3.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.3.1.3 Sampling for testing. Samples for quality conformance testing shall be selected in accordance with inspection level S-3 of MIL-STD-105.

4.3.2 Quality conformance examination.

4.3.2.1 Acceptable quality level. Each sample selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as shown in table I.

Table I. Classification of defects

Categories	Defects	Method of inspection
Major:		
101	Dimensions affecting interchangeability or serviceability not within tolerance.	*S.I.E.
Minor:		
201	Dimensions not affecting interchangeability or serviceability not within tolerance.	*S.I.E.

*S.I.E. - Standard Inspection Equipment.

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4.3.3 Testing.

4.3.3.1 Quality conformance tests. Samples selected in accordance with 4.3.1.3 shall be subjected to the tests specified in table II, using and AQL of 6.5 on the basis of percent defective.

Table II. Applicable tests

Paragraph	Test	First article	Quality conformance	Control
4.5.1.1	Rockwell hardness	X	X	X
4.5.1.2	Density	X	X	X
4.5.1.3	Porosity	X	X	
4.5.1.4	Pressure resistance	X	X	
4.5.1.5	Crush resistance	X	X	
4.5.2	Physical properties of impregnate	X		
4.5.3	Stroking	X		
4.5.4	Corrosion	X		

4.4 Control tests.

4.4.1 Sampling for control tests. Samples for the control tests shall be selected at random and at the rate of 40 pistons per month or 20 of each 10,000 produced, whichever occurs first, except that not more than 80 pistons shall be selected in any 30 day period. Samples shall be subjected to the applicable tests specified in table II.

4.4.2 Failure. Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

4.5 Test procedures.

4.5.1 Piston.

4.5.1.1 Rockwell hardness. Unless otherwise specified, all hardness tests shall be performed on the flat end of the piston and shall be conducted in accordance with ASTM E18.

4.5.1.2 Density. The piston under test shall be coated with a thin layer of petrolatum and the excess removed with a soft cloth. The density of the impregnated piston and the unimpregnated piston shall then be determined by ASTM B328.

4.5.1.3 Porosity. The intercommunicating porosity of the piston shall be determined by ASTM B328. Acetone shall be used as the extracting solvent.

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4.5.1.3.1 Profilograph. The roughness profile of the piston shall be determined by a stylus type surface analyzer using a tracer with a tip radius 0.0005 inch. The vertical sensitivity shall be no less than 100 microinches per 1/4 inch division. The horizontal sensitivity shall be no less than 0.020 inch per 1/2 inch division. The profile trace shall be taken on curved outer surface of the piston, parallel to the center line of the piston.

4.5.1.4 Pressure resistance. A complete wheel cylinder assembly containing the pistons under test shall be completely filled with a suitable hydraulic brake fluid so that no air is entrapped inside the cylinder. Pistons shall be held in place by a suitable clamp. The bleeder valve opening and fluid line connector opening shall be sealed. The cylinder shall be placed on end on a test stand. A push rod having same contact surface dimensions as the push rod used with the actual piston shall be centered and held in place by suitable means parallel to the piston wall. A load shall be applied to the push rod in a direct line by means of a universal testing machine so that the internal fluid pressure on the cylinder attains 3000 $\pm$ 30 psi for 30 seconds. The wheel cylinder shall be disassembled and the pistons examined for compliance with 3.4.1.4. A second cylinder shall be tested with a load applied so that the internal fluid pressure on the cylinder attains 5000 $\pm$ 50 psi for 30 seconds. This cylinder shall be disassembled and the pistons examined for compliance with 3.4.1.4. An optional test method is to locate the pistons in the cylinder above and then pressurize the cylinder by an external means through the inlet hole.

4.5.1.5 Crush resistance. The radial crush resistance of the piston under test shall be determined by means of a universal testing machine equipped with a self-centering head. The piston shall be loaded in a radial position at a rate of 2000 pounds per minute until failure or maximum crush load is observed. Maximum crush load shall be determined by the drop of beam method or a recorded drop of load.

4.5.2 Physical properties of impregnate.

4.5.2.1 Viscosity. The viscosity of the impregnate shall be determined in accordance with ASTM D445 at 100°F.

4.5.2.2 Specific gravity. The specific gravity of the impregnate shall be determined at 20°/20° C in accordance with ASTM D941.

4.5.2.3 Pour point. The pour point of the impregnate shall be determined in accordance with ASTM D97.

4.5.2.4 Flash point. The flash point of the impregnate shall be determined in accordance with ASTM D92.

4.5.3 Stroking (endurance). The stroking properties of the pistons shall be determined in accordance with procedure A of method 361 of Fed. Test Method Std. No. 791. The pistons shall be subjected to 70,000 strokes at 248°F (120°C) and 1000 psi pressure. The test fluid shall be the fluid specified in SAE Standard J1703 (see 6.3).

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4.5.4 Corrosion resistance. Three complete cylinder assemblies of the proper size for the pistons being tested shall be disassembled and cleaned of all deposits, stain, or corrosion. The cylinders shall be reassembled, each part being dipped in the brake fluid specified in SAE Standard J1703 (see 6.3) to which has been added 2 ± 0.1 percent by weight of benzoyl peroxide, reagent grade, and 5 ± 0.1 percent by volume of distilled water. The cylinders shall be reassembled without boots. The pistons shall be held with suitable clamps so that the outside edge of the piston is flush with the outside end of the cylinder bore. The cylinder shall be filled with the brake fluid mixture and placed in a humidity cabinet at 115°F (46°C) and 95 percent relative humidity for 14 days. After the 14 day period, the cylinders shall be disassembled and examined to determine compliance with 3.5.2.

4.6 Inspection of preparation for delivery.

4.6.1 Materials and processes. The Government inspector shall, at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery, to determine conformance to the requirements of section 5 and specifications referenced therein. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1. Preservation, packaging and packing. Preservation, packaging, and packing shall be in accordance with the technical requirements specified on the applicable packaging standard or data sheet and the specified level of preservation, packaging and packing (see 6.2).

6. NOTES

6.1 Intended use. The sintered iron pistons covered by this specification are intended for use in wheel cylinders of hydraulic brake systems and are not intended to be procured as a separate item of supply.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Where first article samples are to be sent (see 4.2).
c. Selection of applicable levels of preservation, packaging, and packing (see 5.1).
d. Selection of applicable packaging standard or packaging data sheet (see 5.1.1.1).

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6.3 Test materials. The standard test brake fluid (RM-1) can be obtained from the Society of Automotive Engineers, Inc., 485 Lexington Avenue, New York, N. Y. 10017.

Custodians:
Army - AT
AF -84

Preparing activity:
Army - AT

Project No. 2530-0239

Review activity:
AF - 84
MR

User activities:
Army - ME, SM

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